On Hilbert's Basis Theorem

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Noetherian rings appear very naturally in the study of rings. The most natural example of a ring one can think of is (perhaps) $\mathbb{Z}$, where every ideal turns out to be generated by a single element. This is an example of a Principal Ideal Domain (PID). But this is too strong of a condition, and oftentimes the rings we care about doesn't have this property. For example, in algebraic geometry, one studies the ring $k[x]$ or $k[x_1, \ldots, x_n]$ where k is a field. Turns out $k[x]$ is a PID, but if we increase the number of variables in the polynomial, $k[x_1, \ldots, x_n]$ is **NOT** a PID for $n \geq 2$. The next best thing we can hope for is that every ideal is **finitely** generated—instead of singly generated. This is precisely the definition of a Noetherian ring.

Hilbert's Basis Theorem states that if R is a Noetherian ring, then so is $R[x]$. This proves that $k[x_1, \ldots, x_n]$ is Noetherian for every $n \geq 1$. This gives a very convenient way of working with the ideals of $k[x_1, \ldots, x_n]$, since they are all finitely generated! In this note, we'll define Noetherian rings and state some of their properties, and we'll prove Hilbert's Basis Theorem, and finally we'll wrap up with how this theorem aids working in algebraic geometry.

§1 Preliminaries

Definition 1.1 (Poset). A **partially ordered set** (or **poset**) is a set X equipped with a binary relation $\leq$ such that

1. $\leq$ is reflexive (i.e. $x \leq x$);
2. $\leq$ is transitive (i.e. $x \leq y, y \leq z$ implies $x \leq z$);
3. $\leq$ is antisymmetric (i.e. $x \leq y$ and $y \leq x$ implies $x = y$).

For $S \subseteq X$, a **maximal element** of S is an element $m \in S$ such that $m \leq s$ for $s \in S$ implies $m = s$.

Proposition 1.1

Let $(X, \leq)$ be a poset. Then the following are equivalent:

1. Every increasing sequence $x_1 \leq x_2 \leq \cdots$ in X is stationary, i.e. there exists n such that $x_n = x_{n+1} = \cdots$.
2. Every non-empty subset of X has a maximal element.

Proof. **(2⇒1)** The set $\{x_1, x_2, \ldots\}$ has a maximal element, say x_n. Then $x_n = x_{n+1} = \cdots$.

(1⇒2) Assume for the sake of contradiction that there exists some set S with no maximal element. Pick $x_1 \in S$. Since it's not maximal, we can pick $x_2 \in S$ such that $x_1 \leq x_2$ and $x_2 \neq x_1$. Again, since S is not maximal, we can pick $x_3 \in S$ such that $x_2 \leq x_3$ and $x_3 \neq x_2$. Thus we can continue forever, and construct an increasing sequence

$$x_1 \leq x_2 \leq \cdots \tag{1.1}$$

that's not stationary. ■

Given a commutative unital ring R (from now on, all rings will be commutative unital, unless mentioned specifically), we have a poset of all ideals of R with set inclusion.

Definition 1.2 (Noetherian ring)**.** A ring R is called **Noetherian** if every increasing chain of ideals

$$I_1 \subseteq I_2 \subseteq I_3 \subseteq \cdots \tag{1.2}$$

eventually stabilizes, i.e. $I_n = I_{n+1} = \cdots$ for some n.

Clearly, every field k is a Noetherian ring since the only ideals are (0) and k itself.

Proposition 1.2

R is Noetherian if and only if every ideal is finitely generated.

Proof. **(⇒)** Assume for the sake of contradiction that there is an ideal I which is not finitely generated. Pick any $r_1 \in I$. We pick r_{n+1} as follows: since $(r_1, \ldots, r_n)$ isn't the whole I, so we can find some $r_{n+1} \in I \setminus (r_1, \ldots, r_n)$. This gives rise to an increasing chain of ideals:

$$(r_1) \subsetneq (r_1, r_2) \subsetneq (r_1, r_2, r_3) \subsetneq \cdots \tag{1.3}$$

and it never stabilizes. Contradiction!

(⇐) Suppose every ideal of R is finitely generated. Let's take an increasing chain of ideals:

$$I_1 \subseteq I_2 \subseteq I_3 \subseteq \cdots \tag{1.4}$$

Consider the ideal $I = \bigcup_{i=1}^{\infty} I_i$. By hypothesis, it is finitely generated, i.e.

$$\bigcup_{i=1}^{\infty} I_i = (r_1, \ldots, r_m)\,. \tag{1.5}$$

Then there is some ideal I_n that contains all the generators r_i. Then we'll have $I_n = I_{n+1} = I_{n+2} = \cdots$, and hence R is Noetherian. ■

Remark 1.1. Mimicking the proof of the forward direction, one can show that if an ideal $I = (G)$ is generated by a generating subset G, then there is a finite subset of G that generates I. Indeed, if no finite subset of G generates I, then starting from $r_1 \in G$, we can pick r_{n+1} as follows: since no finite subset of G generates I, $(r_1, \dots, r_n)$ isn't the whole I, so we can find some $r_{n+1} \in I \setminus (r_1, \dots, r_n)$ with $r_{n+1} \in G$. This gives rise to an increasing chain of ideals:

$$(r_1) \subsetneq (r_1, r_2) \subsetneq (r_1, r_2, r_3) \subsetneq \cdots \tag{1.6}$$

and it never stabilizes.

§2 Hilbert's basis theorem and its proof

Theorem 2.1 (Hilbert's Basis Theorem)

If R is Noetherian, so is $R[x]$.

Proof. Assume for the sake of contradiction that $R[x]$ is not Noetherian. So it has an ideal $\mathfrak{I}$ that is not finitely generated. Pick $f_1 \in \mathfrak{I}$ of minimal degree. We pick f_{n+1} as follows: since $(f_1, \dots, f_n)$ isn't the whole $\mathfrak{I}$, we can pick $f_{n+1} \in \mathfrak{I} \setminus (f_1, \dots, f_n)$ of minimal degree. This gives rise to an increasing chain of ideals:

$$(f_1) \subsetneq (f_1, f_2) \subsetneq (f_1, f_2, f_3) \subsetneq \cdots \tag{2.1}$$

By construction,

$$\deg f_1, \deg f_2, \deg f_3, \dots \tag{2.2}$$

is a non-decreasing sequence of non-negative integers. Now, let a_n be the leading coefficient of f_n. Then we form the ideal

$$\mathfrak{a} = (a_1, a_2, a_3, \dots) \trianglelefteq R. \tag{2.3}$$

Since R is Noetherian, this is finitely generated. By Remark 1.1, there exists some N such that

$$\mathfrak{a} = (a_1, a_2, a_3, \dots, a_N) \trianglelefteq R. \tag{2.4}$$

In particular,

$$a_{N+1} = \sum_{i=1}^{N} r_i a_i \tag{2.5}$$

for $r_i \in R$. Now, consider

$$g = \sum_{i=1}^{N} r_i \; x^{\deg f_{N+1} - \deg f_i} \; f_i \in (f_1, \dots, f_N)\,. \tag{2.6}$$

Its leading term is $\sum_{i=1}^{N} r_i a_i = a_{N+1}$, which is the leading term of f_{N+1}. Therefore, $f_{N+1} - g$ has degree less than f_{N+1}, and it's not in $(f_1, \dots, f_N)$. This contradicts the minimality of $\deg f_{N+1}$, since f_{N+1} was choesn to be the minimal-degree polynomial in $\mathfrak{I} \setminus (f_1, \dots, f_N)$.

Thus we arrived at a contradiction, starting from the assumption that $R[x]$ has a non-finitely generated ideal $\mathfrak{I}$. Therefore, this assumption must be false, i.e. $R[x]$ is Noetherian. ■

Corollary 2.2

If R is Noetherian, so is $R[x_1, x_2, \ldots, x_n]$.

Proof. This can be seen by induction on n. The base case $n = 1$ is just Hilbert's Basis Theorem. Inductively, suppose $R[x_1, x_2, \ldots, x_{n-1}]$ is Noetherian. Then since

$$R[x_1, x_2, \ldots, x_n] = R[x_1, x_2, \ldots, x_{n-1}][x_n], \tag{2.7}$$

by Hilbert's Basis Theorem, $R[x_1, x_2, \ldots, x_n]$ is Noetherian. ■

In particular, if k is a field, $k[x_1, \ldots, x_n]$ is Noetherian. However, $k[x_1, x_2, \ldots]$ is not Noetherian, since the chain of ideals

$$(x_1) \subseteq (x_1, x_2) \subseteq (x_1, x_2, x_3) \subseteq \cdots \tag{2.8}$$

never stabilizes.

§3 What's so special about it?

The significance of Hilbert's Basis Theorem is best understood if it's considered in the context of algebraic geometry. In algebraic geometry, we have the notion of a variety, which is a common zero-set of polynomials. Formally speaking,

Definition 3.1 (Affine algebraic set)**.** Let $\Bbbk$ be an algebraically closed field, and let $\mathbb{A}^n$ be the set of all n-tuples of elements of $\Bbbk$. For $S \subseteq \Bbbk[x_1, \ldots, x_n]$, define the zero-locus $Z(S)$ to be

$$Z(S) = \{P \in \mathbb{A}^n \mid f(P) = 0 \text{ for all } f \in S\}. \tag{3.1}$$

A subset $V \subseteq \mathbb{A}^n$ is called an **affine algebraic set** if $V = Z(S)$ for some $S \subseteq \Bbbk[x_1, \ldots, x_n]$.

The main objects of interest in algebraic geometry are these affine algebraic sets. We declare them to be the closed sets, and thus form the Zariski topology on $\mathbb{A}^n$. These geometric objects are defined by a system of polynomial equations. But, in principle, there might be infinitely many polynomial equations, which might be very messy to deal with. But thanks to Hilbert's Basis Theorem, such a system can always be described using only a finite number of polynomial equations!

Proposition 3.1

For $S \subseteq \Bbbk[x_1, \ldots, x_n]$, $Z(S) = Z(\langle S \rangle)$.

Proof. Since $S \subseteq \langle S \rangle$, it's clear that $Z(S) \supseteq Z(\langle S \rangle)$. Now, for $P \in Z(S)$, then $f(P) = 0$ for all $f \in S$. Then for any $\sum_{i=1}^{n} f_i g_i \in \langle S \rangle$ with $f_i \in S$,

$$\left(\sum_{i=1}^{n} f_i g_i \right)(P) = \sum_{i=1}^{n} f_i(P) g_i(P) = 0. \tag{3.2}$$

Therefore, $P \in Z(\langle S \rangle)$, proving that $Z(S) \subseteq Z(\langle S \rangle)$. ∎

Since $\Bbbk[x_1, \ldots, x_n]$ is Noetherian, thanks to Hilbert's Basis Theorem, any such ideal $\langle S \rangle$ is generated by finitely many polynomials, say $f_1, \ldots, f_m$. Then we can write

$$Z(S) = Z(\langle S \rangle) = Z(\langle f_1, \ldots, f_m \rangle) = Z(\{f_1, \ldots, f_m\}). \tag{3.3}$$

Hence, any algebraic set is a common zero-set of just finitely many polynomials! The set S can be as large as it can get, it can be uncountable as well. But there is always a finite collection of polynomials such that the zero-set of S is precisely the common zero-set of those finitely many polynomials.

Also, Hilbert's Basis Theorem is a crucial stepping stone for major theorems in algebraic geometry, like Hilbert's Nullstellensatz, which connects geometric objects (zero-loci) directly to algebraic objects (ideals).

Although it should be noted that the proof of Hilbert's Basis Theorem is non-constructive. So, the proof doesn't say anything on how to construct those finitely many polynomials. But it laid the foundational framework for modern computer algebra systems that use tools like Gröbner bases to solve systems of polynomial equations. In fact, there's an algorithm called Buchberger's algorithm that computes Gröbner bases, and the fact that this algorithm always terminates is guaranteed by Hilbert's Basis Theorem.

References

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